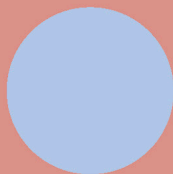
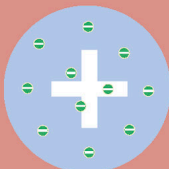


SOLID SPHERE MODEL



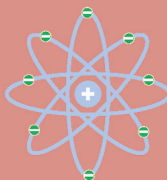
In 1803, John Dalton presented his theory that all matter is made from atoms—indivisible spheres that cannot be created or destroyed. However, atoms can be bonded or broken apart from other atoms to form new substances.

PLUM PUDDING MODEL



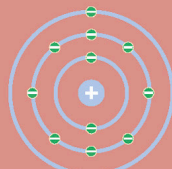
In J.J. Thomson's model, negatively charged electrons are dotted randomly throughout a sphere, which has a positive charge.

RUTHERFORD MODEL



Experimental evidence led Ernest Rutherford in 1911 to propose that the entire positive electric charge in an atom lay in a small, dense core and the electrons were imagined to orbit around this nucleus, like moons around a planet.

BOHR MODEL



To explain light absorption and emission by atoms, Niels Bohr developed a model in which electrons could orbit only in particular energy "shells."

THREE TINY PIECES

Atoms are the fundamental building blocks of large-scale matter—particles that were first thought indivisible and whose collective chemical and physical properties make them representative of one or another specific element. On a deeper level, however, all atoms are made up of a combination of three subatomic particles: positively charged protons and uncharged neutrons in a central nucleus, and negatively charged electrons orbiting in more distant clouds (see p.31), which allow atoms to bond with other atoms.